

## PS-05: Personal Finance Agent

**Domain: FinTech**

Build an AI financial assistant that analyzes bank statements, categorizes spending, detects anomalies, tracks budgets, and provides proactive financial insights and alerts.

**In essence:**

An always-on personal finance advisor that monitors spending and improves financial health.

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## Build an Autonomous Financial Wellness Agent

Develop a multi-agent AI system capable of:

- Understanding financial transactions.
- Categorizing expenses automatically.
- Tracking spending patterns.
- Detecting unusual activity.
- Monitoring budgets.
- Forecasting future cash flow.
- Generating personalized financial advice.
- Alerting users before financial problems occur.

The system should function as a continuous financial advisor rather than a passive budgeting application.

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## Example Scenario

### User Uploads

3 Months Bank Statements

or

CSV

PDF  
UPI Transaction History  
Credit Card Statements

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## Traditional Finance App

Food: ₹12,000

Travel: ₹3,000

Shopping: ₹6,000

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## Agentic Finance System

Detected:

Food expenses increasing 18% monthly.

3 unused subscriptions found.

Electricity bill unusually high.

Projected savings this month:  
₹6,500

Potential annual savings:  
₹22,000

The system actively improves financial health.

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## Core Components

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### 1. Financial Data Ingestion Agent

First responsibility:

Collect and normalize financial data.

Sources:

Bank Statements  
Credit Cards  
UPI Transactions  
Wallets  
Investment Accounts

Example:

HDFC Statement.pdf

Converted into:

```
{  
  "date": "2026-02-15",  
  "amount": 250,  
  "merchant": "Swiggy"  
}
```

All data becomes structured.

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## 2. Transaction Categorization Agent

Automatically classifies spending.

Example:

Swiggy

↓

Food & Dining

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Uber

↓

Transportation

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Netflix

↓

Entertainment

Possible categories:

Food

Travel

Healthcare

Shopping

Bills

Education

Investments

Subscriptions

The system continuously improves classification accuracy.

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### 3. Spending Pattern Analysis Agent

This is where intelligence begins.

Example:

Food Spending

January: ₹5,000

February: ₹6,500

March: ₹8,000

Agent detects:

Consistent increase trend.

Another example:

Fuel Spending

Usually: ₹2,000

Current Month: ₹6,500

Flag anomaly.

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## 4. Budget Management Agent

Creates and manages budgets.

Example:

Monthly Income

₹50,000

Recommended:

Needs: ₹25,000

Wants: ₹15,000

Savings: ₹10,000

The budget evolves with spending behavior.

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## 5. Financial Health Scoring Agent

Think of this as:

Credit Score

+

Budget Discipline

+

Savings Consistency

Combined into:

Financial Health Score

82 / 100

Factors:

- Savings rate
- Debt ratio
- Spending consistency
- Emergency fund coverage

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## 6. Anomaly Detection Agent

One of the strongest features.

Example:

Normal Spending:  
₹500/day

Suddenly:

₹15,000 transaction

Agent flags:

Potential Fraud

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Another example:

Netflix billed twice

↓

Duplicate Charge Detected

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## 7. Subscription Intelligence Agent

Most users forget subscriptions.

Detect:

Netflix  
Spotify  
Amazon Prime  
ChatGPT

Identify:

Unused subscriptions

Example:

Spotify subscription active

No transactions or usage indicators for 4 months

Recommendation:

Cancel

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## 8. Cash Flow Forecasting Agent

Very valuable for demos.

Based on:

Income

Bills

EMIs

Recurring Expenses

Predict:

Expected Balance

Example:

Current Balance:

₹32,000

Predicted Balance after 25 days:

₹4,500

Alert user before financial stress occurs.

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## 9. Financial Advisor Agent

This is the most "agentic" component.

Instead of reporting:

Spent ₹10,000 on food

Recommend:

Reducing restaurant visits by 2 per week could save approximately ₹18,000 annually.

Another example:

Emergency fund covers only 1.2 months.

Recommended:

Build reserve for 6 months.

The system becomes a financial coach.

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## 10. Alert & Notification Agent

Provides proactive alerts.

Examples:

### **Budget Risk**

⚠ Food budget 90% utilized.

### **Fraud Alert**

⚠ Unusual spending detected.

### **Subscription Alert**

⚠ Unused subscription detected.

### **Savings Alert**

⚠ Savings rate dropped below target.

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## Multi-Agent Architecture

A serious implementation might include:

### **Agent 1**

Data Ingestion Agent

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## **Agent 2**

Transaction Categorization Agent

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## **Agent 3**

Spending Analysis Agent

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## **Agent 4**

Budget Planning Agent

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## **Agent 5**

Financial Health Agent

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## **Agent 6**

Anomaly Detection Agent

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## **Agent 7**

Subscription Intelligence Agent

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## **Agent 8**

Cash Flow Forecast Agent

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## **Agent 9**

Financial Advisor Agent

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## Agent 10

Notification Agent

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## Agent 11

Master Orchestrator

Coordinates the entire workflow.

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# Advanced Features (What Can Win)

## 1. Goal-Based Planning

User sets:

Buy Car in 2 Years

Agent calculates:

Need to save ₹8,500/month.

Tracks progress automatically.

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## 2. Life Event Simulation

Example:

What if I buy a ₹12 lakh car?

Agent simulates:

EMI Impact

Savings Impact

Cash Flow Impact

Very impressive in demos.

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### **3. Financial Twin**

Create a digital representation of the user's finances.

The agent can test:

New House  
New Loan  
Salary Increase  
Job Loss

and predict outcomes.

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### **4. Investment Readiness Agent**

Instead of recommending specific investments, assess readiness.

Example:

Emergency Fund:  
Incomplete

Recommendation:  
Complete emergency savings before increasing investment risk.

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### **5. Family Finance Dashboard**

Track:

Parents  
Spouse  
Children

Combined financial health.

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## 6. AI CFO Dashboard

Display:

Financial Health Score

Monthly Savings

Budget Utilization

Fraud Alerts

Cash Flow Forecast

Upcoming Bills

Excellent judging demo.

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## What Judges Will Likely Reward

Not:

Statement



Categorization



Charts

But:

Financial Data



Understand Spending



Detect Patterns



Predict Risks



Recommend Actions



Monitor Continuously



Improve Financial Health